

	Course Name:	Electronic Devices & Circuits	Course Code:	EE1004
	Program:	Electrical Engineering	Semester:	Fall 2022
	Duration:	60 Minutes	Total Marks:	50
	Paper Date:	12-11-2022	Weight	15%
	Section:	ALL	Page(s):	8
Exam Type:	Mid-2			

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Section: B6C

Instruction/Notes:

- Carefully read the paper and attempt each question only in the space provided.
- Show all calculations, direct answers are not acceptable.
- State your valid assumptions clearly, if you have to take any.

-3A

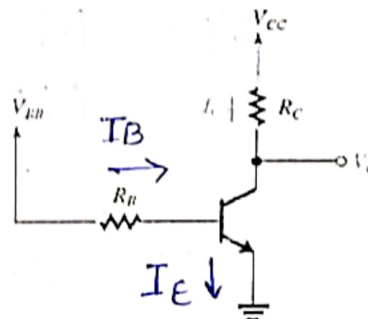
	Question 1 (CLO#2)	Question 2 (CLO#2)	Total
Total marks	20	30	50
Marks obtained	<u>20</u>	<u>15</u>	<u>35</u>
CLO score			<u>(4)</u>

Question: 1 (CLO # 2)

20 Marks

For the circuit shown in figure, $V_{CC} = 12V$, $R_C = 4.7k\Omega$ and $R_B = 150k\Omega$.

The BJT has $\beta = 150$



Find the value of V_{BB} that result in transistor operating

- In active mode with $V_C = 6V$
- At the edge of saturation
- Deep in saturation with $\beta_{forced} = 15$.

$V_{CC} = 12V$, $R_C = 4.7K\Omega$, $R_B = 150K\Omega$, $\beta = 150$

a) $V_C = 6V$

finding I_C :

$$I_C = \frac{V_{CC} - V_C}{R_C}$$

$$I_C = \frac{12 - 6}{4.7 \times 10^3}$$

$$I_C = 1.27 \text{ mA}$$

Now;

$$I_B = \frac{I_C}{\beta}$$

$$I_C = \beta I_B$$

$$I_B = \frac{1.27 \times 10^{-3}}{150}$$

$$I_B = 8.46 \times 10^{-6} \text{ A}$$

In Active Mode:-

$$V_{BE} = 0.7$$

$$V_B - V_E = 0.7$$

$$V_E = 0$$

$$\Rightarrow V_B = 0.7$$

V_{BB} :-

$$I_B = \frac{V_{BB} - V_B}{R_B}$$

$$I_B R_B = V_{BB} - V_B$$

$$I_B R_B + V_B = V_{BB}$$

$$1.27 + 0.7 = V_{BB}$$

$$V_{BB} = 1.97 \text{ V}$$

b) at edge of saturation:-

$$V_{CE} = 0.3$$

$$V_C - V_E = 0.3 \text{ V}$$

$$V_C = 0.3 \text{ V} \rightarrow \text{EOS}$$

$$I_C = \frac{V_{CC} - V_C}{R_C} = 2.48 \times 10^{-3} \text{ A}$$

finding:-

$$I_B = ? \rightarrow 150$$

$$I_C = \beta I_B$$

$$I_B = 1.65 \times 10^{-5} \text{ A}$$

where,

$$V_B = 0.7$$

finding V_{BB} :-

$$I_B = \frac{V_{BB} - V_B}{R_B}$$

$$I_B R_B + V_B = V_{BB}$$

$$V_{BB} = (1.65 \times 10^{-5}) (150 \times 10^3) + 0.7$$

$$V_{BB} = 3.175 \text{ V}$$

c) $\beta_f = 150$

$$V_{CE(\text{sat})} = 0.2 \text{ V}$$

$$V_C(\text{sat}) - V_E(\text{sat}) = 0.2$$

$$V_C(\text{sat}) = 0.2 \text{ V}$$

finding I_C

$$I_C = \frac{V_{CC} - V_C(\text{sat})}{R_C}$$

~~$I_C = 2.1$~~
Now;

$$\boxed{I_C = 2.51 \text{ mA}}$$

finding I_B

$$I_C = \beta_f I_B$$

$$I_B = \frac{2.51 \times 10^{-3}}{15}$$

$$I_B = 1.66 \times 10^{-4} \text{ A}$$

Now finding $V_B = ?$

$$V_{BE} = 0.7$$

$$V_B - V_E = 0.7$$

$$\boxed{V_B = 0.7}$$

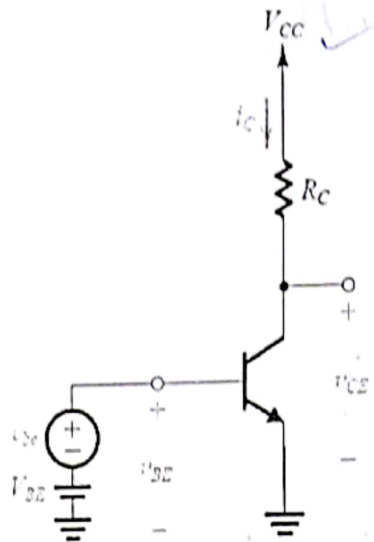
finding V_{BB} :

$$V_{BB} = I_B(\text{sat}) R_B + V_B$$

$$V_{BB} = (1.66 \times 10^{-4})(150 \times 10^3) + 0.7$$

$$V_{BB} = 25.6 \text{ V}$$

Consider the circuit shown in the figure below operated with a supply voltage $V_{CC} = +4V$.



- a) What is the theoretical maximum voltage gain that this amplifier can provide?

$$\rightarrow \text{Voltage gain} = - \frac{(V_{CC} - V_{CE})}{V_T}$$

$\Rightarrow$

$$\text{Voltage gain} = -71.2 \text{ when } V_{CE} = 2.22V$$

why have you taken $V_{CE} = 2.2$?

- b) What value of V_{CE} must this amplifier be biased at to provide a voltage gain of $-70 V/V$?

$$V_{CC} = +4$$

$$\text{where } V_E = 0$$

$$V_{CE} = ?$$

when

$$V_{CE} = 2.22V$$

$$V_{BE} = 0.7$$

$$V_E = 0$$

$$V_B = 0.7$$

$$\text{Voltage gain} = - \frac{(V_{CC} - V_{CE})}{V_T}$$

$$= - \frac{(4 - 2.22)}{25 \times 10^{-3}}$$

$$= -71.2$$

- c) If the dc collector current I_C at the bias point in (b) is to be 0.5 mA, what value of R_C should be used?

$$I_C = 0.5 \text{ mA}$$

$$V_{CC} = 4 \text{ V}$$

where $V_{CE} = 2.2 \text{ V}$

$$V_C - V_E = 2.2 \text{ V}$$

$$\boxed{V_C = 2.2 \text{ V}}$$

$$\frac{V_{CC} - V_C}{R_C} = I_C$$

$$R_C = \frac{V_{CC} - V_C}{I_C}$$

$$R_C = \frac{4 - 2.2}{0.5 \times 10^{-3}}$$

$$R_C = 3.6 \text{ k}\Omega$$

- d) What is the value of V_{BE} required to provide the bias point mentioned above? Assume that the BJT has $I_S = 10^{-15} \text{ A}$.

as $I_C = I_S e^{\frac{V_{BE}}{V_T}}$

$$0.5 \times 10^{-3} = 1 \times 10^{-15} e^{\frac{V_{BE}}{25 \times 10^{-3}}}$$

$$5 \times 10^{11} = e^{\frac{V_{BE}}{25 \times 10^{-3}}}$$

$$\ln(5 \times 10^{11}) = \frac{V_{BE}}{25 \times 10^{-3}}$$

$$26.93 = \frac{V_{BE}}{25 \times 10^{-3}}$$

$$\boxed{V_{BE} = 0.67 \text{ V}}$$

- e) If a sine-wave signal v_{be} having a 6-mV peak amplitude is superimposed on V_{BE} , find the corresponding output voltage signal v_{ce} that will be superimposed on V_{CE} assuming linear operation around the bias point.

$$\rightarrow \boxed{v_{be} = 6 \text{ mV}}$$

$$\boxed{V_{CE} = 2.22 \text{ V}}$$

$$\Rightarrow \text{where } V_{BE} = 0.67$$

$$V_{BE} = V_{BE} + v_{be}$$

$$V_{BE} = 0.67 + 6 \times 10^{-3}$$

$$\boxed{V_{BE} = 0.676} \quad \checkmark$$

$$\text{where } g_m = \frac{I_C}{V_T}$$

$$g_m = \frac{0.5 \times 10^{-3}}{25 \times 10^{-3}}$$

$$\boxed{g_m = 0.02}$$

$$i_c = g_m \times v_{be}$$

$$v_{ce} = ?$$

(ac)

- f) What is the value of the dc base current I_B at the bias point? Assume $\beta = 100$. Find the ac signal current i_b that will be superimposed on the base current I_B .

$$\beta = 100$$

$$I_B = ?$$

where,

$$I_C = 0.5 \text{ mA}$$

$$\Rightarrow I_C = \beta I_B$$

$$I_B = \frac{0.5 \times 10^{-3}}{100}$$

$$\rightarrow \boxed{I_B = 5 \times 10^{-6} \text{ A}}$$

$$\boxed{i_B = I_B + i_b}$$

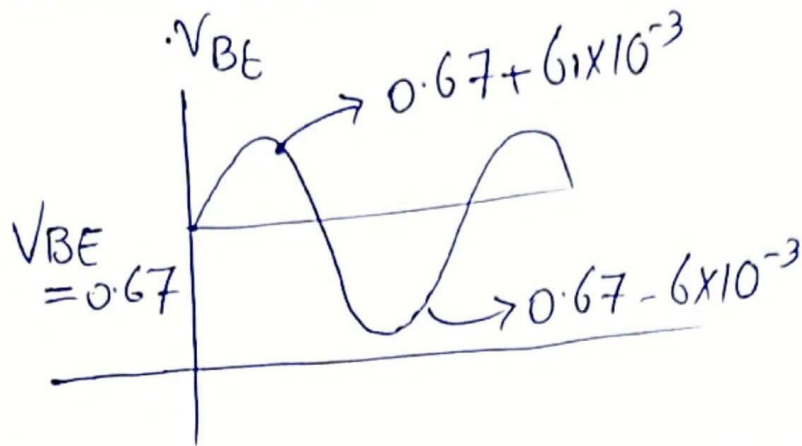
where

$$v_{be} = 6 \text{ mV}$$

$$V_{BE} = 0.676$$

$$\boxed{g_m = \frac{i_c}{v_{be}}}$$

- g) Sketch and clearly label correlated graphs for v_{BE} , v_{CE} , i_C , and i_B . Note that each graph consists of a dc or average value and a superimposed sine wave. Be careful of the phase relationships of the sine waves.

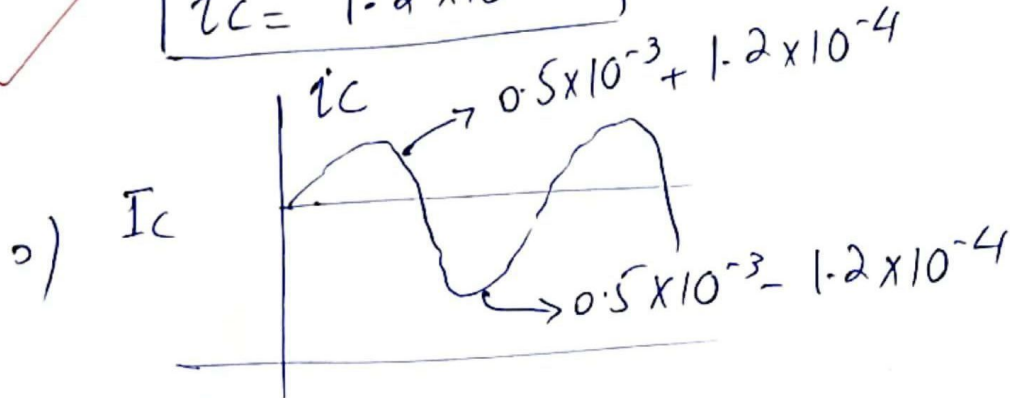


$$i_C = g_m v_{be}$$

$$i_C = g_m$$

$$i_C = 0.02 \times 6 \times 10^{-3}$$

$$i_C = 1.2 \times 10^{-4} A$$



o) 4

$$\frac{15}{30}$$